

Security-Aware Workload-State Counterfactual Verification and Nodal Net-Value Settlement for Spatially Coupled AI Data Centers

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Abstract—Artificial-intelligence (AI) data centers can provide demand response (DR), but meter decreases do not show pre-event workload commitment or positive network value. Delayed service and cross-site rebound make workload-feasibility and signed-value checks essential. This paper develops a declaration-indexed verifier for spatially coupled data centers. Submit-time scheduler records, including release, declared runtime, graphics processing unit (GPU) request, class, and region, define a cumulative workload-state model, a convex risk contract, and a contiguous runtime witness on one ledger. An active-window energy map includes carry-in jobs, while execution telemetry remains outside decision-time fitting. The risk fit combines total exposure with conditional value-at-risk (CVaR) under a validation-frozen mean-square-error (MSE) bound; exact job-start enumeration audits its finite executable bridge. In a 54-day mechanism-isolation panel, the joint fit reaches normalized root-mean-square error (nRMSE) 0.902, false credit 0.780 MWh/day, under-credit 26.506 MWh/day, and an F1 score (harmonic mean of precision and recall) of 0.536. The total-budget-only fit has lower nRMSE, false credit, and under-credit, higher F_1 (0.549 versus 0.536), and lower false-credit CVaR. Thus, these locked results do not establish an empirical benefit from the tail term. At the slot-62 information gate, nRMSE is 1.366 and F_1 is 0.080. The job-level bridge exactly audits 7,306,622 feasible starts, while its flexible-load relative- L_2 residual is 1.581. The IEEE Reliability Test System 24-bus (RTS-24) result is a declaration-only single-line N-1 network replay; payment activation requires a closed meter, and observational workload traces do not identify a utility intervention effect.

Index Terms—AI data centers, counterfactual verification, demand response, security-constrained dispatch, nodal settlement.

I. INTRODUCTION

The growth of large-model training and inference is converting data centers from nearly static industrial loads into large, software-defined grid resources. Interactive inference is latency constrained, elastic inference can be delayed briefly, and batch GPU work can be deferred for hours or migrated between regions. These capabilities enable demand response (DR), but they also invalidate the usual interpretation of a meter decrease. A reduction at one facility can be followed by delayed consumption or simultaneous rebound at another node. Paying the positive local decrease can therefore reward a redistribution that creates little or negative network value. Public traces already expose multi-week bursty workload arrivals and tens of thousands of scheduler jobs, while facility telemetry reports GPU energy at a different measurement boundary [1], [2].

Conventional DR measurement and verification estimates the unobserved no-event demand from historical meter data.

Arithmetic historical baselines remain common in market practice under published market guidance [3], despite known bias and manipulation incentives [4]. Regression-based facility models quantify baseline uncertainty but do not encode a computing workload ledger [4]. More broadly, DR research establishes the economic role of responsive load, yet a meter-only estimator cannot determine whether queued jobs could actually have produced its predicted trajectory.

Data-center scheduling studies provide this physical state: temporal deferral and geographical balancing reduce energy cost and peak demand [5]. Data-driven flexibility and security-constrained coordination optimize spatial-temporal dispatch [5], [6]. Balancing-market and load-shifting studies provide complementary dispatch contexts. Virtual-link clearing and space-time remuneration model geographical and temporal transfers [7], [8]. Receding-horizon frequency regulation retains multi-site state and future arrivals [9]. These methods optimize operator-controlled schedules; verification of an independently submitted DR counterfactual against an auditable workload ledger remains open. Coupled-regulation models likewise provide no provenance-bound certificate [10]. Whole-facility measurements also require a bottom-up GPU-to-facility conversion; calibration, benchmark scaling, and network placement are therefore kept as separate evidence layers.

The second gap is electrical. Geographic workload control is often valued with energy prices or gross megawatts, although nodal marginal value changes when congestion or a contingency binds. Direct-current (DC) power flow and linear distribution factors provide transparent sensitivities [11], [12]; N-1 dispatch imposes credible post-contingency limits through line-outage distribution factors (LODFs) [13]. No data-center DR verifier jointly certifies workload feasibility, signed spatial response, and security-aware network value. Alternating-current (AC) feasibility is a separate nonlinear check. Public traces support observational replay but not a causal utility event, so measured execution, counterfactual construction, and network valuation must remain separate.

This paper makes three power-system contributions. 1) It defines a typed submit-execute-settle state in which a runtime-complete job witness is formed from release, declared runtime, GPU request, and regional assignment. An active-window energy map gives the exact declaration energy intersecting a locked operating window, including carry-in jobs. The same indexed digest is rechecked at regional aggregation and

reused by the RTS-24 N-1 value replay. 2) It formulates a workload-state verifier with a causal information boundary, a validation-selected high-5-of-10 Pareto projection, and an exact convex total-plus-daily-CVaR contract. A predeclared 10% MSE neighborhood and a triangle-inequality certificate prevent risk reduction from being purchased by an unreported accuracy change; the full simplex is solved numerically and audited with Karush–Kuhn–Tucker (KKT) and primal residuals. The aggregate contract is then mapped to a fixed linear dual-price surrogate and an exact finite contiguous-block policy; the resulting target residual is reported rather than assumed away. 3) It gives the workload response a signed nodal value under complete finite N-1 contingencies and separates declaration feasibility, network-value replay, and meter-gated payment activation. This separation prevents a declaration trajectory from being interpreted as a closed-meter payment. The resulting coupling invariant carries one declared workload state into physical verification and security-aware valuation; its LP, dispatch, and allocation primitives retain established power-system semantics [14]. Because public traces do not identify utility geography, spatial conclusions are conditional on the declared feature-stratified trace scenario and the complete permutation/penetration panel. The certificate assumes trusted scheduler provenance: a digest detects record changes but cannot authenticate semantically false fields.

The remainder of the paper is organized as follows. Section II defines the spatial workload and network settlement problem. Section III presents the verifier, security-aware value rule, and allocation mechanism. Section IV specifies the data, baselines, and locked protocol. Section V reports the core counterfactual, execution-replay, job-level, security, payment, continuous-horizon, and provenance experiments. Section VI concludes with the deployment boundary and future validation requirements.

II. SYSTEM MODEL AND PROBLEM FORMULATION

A. Spatial workload state

Let $\mathcal{S}$, $\mathcal{K}$, $\mathcal{D}$, and $\mathcal{T}$ denote source regions, workload classes, destination data centers, and 15-minute intervals; the symbol h is reserved for a calendar day in statistical panels. The submit-time declaration ledger supplies arrival energy a_{skt} for source s , class k , and release interval t . The decision $x_{skd\tau} \geq 0$ is the energy served at destination d in interval τ . Its cumulative state is

$$y_{skt} = \sum_{\tau=0}^t \sum_{d \in \mathcal{D}} x_{skd\tau}. \quad (1)$$

With cumulative arrivals $A_{skt} = \sum_{\tau=0}^t a_{sk\tau}$ and deadline D_k , release and deadline feasibility are

$$y_{skt} \leq A_{skt}, \quad (2)$$

$$y_{skt} \geq A_{sk,t-D_k}, \quad t \geq D_k. \quad (3)$$

The state recursion is $y_{skt} - y_{sk,t-1} = \sum_d x_{skd\tau}$ with $y_{sk,-1} = 0$, and terminal service equals the declared arrivals due in the window; the release and deadline inequalities are (2) and (3), and the site-capacity constraint is $\sum_{s,k} x_{skd\tau} \leq \bar{P}_d \Delta t$. Facility demand is

$$p_{dt} = P_d^{\text{fix}} + \Delta t^{-1} \sum_{s,k} x_{skd\tau}. \quad (4)$$

For rolling window end T^+ and final event-day release T^0 , the terminal state is not forced to consume unexpired future work. Instead,

$$y_{skT^+} = A_{sk, \max\{T^0, T^+ - D_k\}}, \quad (5)$$

which completes every job due inside the window and every event-day arrival while retaining later real arrivals. The window starts $D_{\max} + 96$ intervals before and ends $D_{\max}$ intervals after the event; omitted earlier work has expired. Equations (2)–(4) map an aggregate preemptive envelope to the facility meter. This is valid for checkpointable batch work; a separate execution audit checks contiguous intervals, runtime, GPU count, and native power. For the counterfactual task-level check, let $j \in \mathcal{J}_{\text{sub}}$ index a submitted scheduler record, r_j denote its submitted release, and let d_j be the service-window endpoint. Slurm “timelimit” is an allocation run-time declaration; it is converted to $R_j = \lceil \text{timelimit}_j / \Delta t \rceil$ slots and combined with a precommitted queue allowance $B = 96$ slots, so $d_j = r_j + R_j + B$. The unlimited Slurm sentinel is mapped to the precommitted horizon $H_{\infty} = 128$ slots before adding B ; finite declarations are retained exactly. The 0.001-MW-per-GPU nameplate is fixed before the split. The training-calibrated entitlement $\hat{E}_j$ is retained only for the aggregate Exp19 audit; the executable witness uses the declared runtime. Let $\bar{p}_{\text{GPU}} = 0.001$ MW/GPU be the committed nameplate, s_j the submit-time scenario region, and g_j^{req} the requested GPU count. Define the runtime-complete entitlement $E_j^{\text{run}} = R_j g_j^{\text{req}} \bar{p}_{\text{GPU}} \Delta t$. The indexed service variable $u_{j\tau}$ is the energy assigned to job j in interval τ , and the executable witness requires one contiguous fixed-rate block of exactly R_j slots. Let w_{batch} be the predeclared waiting-cost coefficient per MWh-slot, and r^{DR} the event service tariff. Define $c_j = g_j^{\text{req}} \bar{p}_{\text{GPU}} \Delta t$ and $\Theta_j = \{r_j, \dots, d_j - R_j\}$. For each admissible start θ , a binary $z_{j\theta}$ selects the block and $u_{j\tau}^{\theta} = c_j \mathbf{1}\{\theta \leq \tau < \theta + R_j\}$. The exact start-time mixed-integer linear program (MILP) is

$$\begin{aligned} \sum_{\theta \in \Theta_j} z_{j\theta} &= 1, \quad z_{j\theta} \in \{0, 1\}, \\ u_{j\tau} &= \sum_{\theta \in \Theta_j} u_{j\tau}^{\theta} z_{j\theta}, \quad \sum_{j: s_j = d} u_{j\tau} \leq \bar{P}_d \Delta t, \\ \min_z \quad &\sum_{j, \theta \in \Theta_j} z_{j\theta} \sum_{\tau} [w_{\text{batch}}(\tau - r_j) + r^{\text{DR}} \mathbf{1}\{\tau \in \mathcal{E}\}] u_{j\tau}^{\theta}. \end{aligned} \quad (6)$$

Every selected block enforces runtime completion, contiguity, and the GPU nameplate; positive regional slack makes the separable enumeration exact for the displayed MILP. For delay $\delta = \theta - r_j$, the waiting part of the objective is $w_{\text{batch}} c_j \{R_j \delta + R_j(R_j - 1)/2\}$, so a long job is charged for every delayed service interval rather than only for its start time. The runtime-complete ledger enumerates every admissible start for all 75,326 records and stores the earliest-start baseline and tariff response; a separate capacity case activates a binding regional row. The measured $[t_j^{\text{start}}, t_j^{\text{end}}]$ interval is opened only for independent post-event scoring after the declaration witness is fixed: 71,128 execution matches never constrain z , u , or d_j . Thus $\hat{E}_j$ is calibration-only, E_j^{run} is the nameplate-capacity commitment, and the calibrated central energy is carried on the same blocks without being presented as measured consumption. The risk target is linked through a fixed regional slot price. Because the

statistical and nameplate ledgers have different scales, define the locked window set $\mathcal{W}_{\text{lock}} = \bigcup_{h \in \mathcal{H}_{\text{lock}}} \{96h, \dots, 96h + 95\}$. The declaration energy active in this set is

$$E_{\text{act}}(\mathcal{W}_{\text{lock}}) = \sum_{j \in \mathcal{J}_{\text{sub}}} E_j^{\text{dec}} \frac{|[r_j, r_j + R_j] \cap \mathcal{W}_{\text{lock}}|}{R_j}, \quad (7)$$

where E_j^{dec} is either the nameplate upper entitlement or the calibrated central entitlement. This overlap integral in (7) includes carry-in jobs and uses only submit-time fields. Let $\tilde{p}_{dht}^{\text{risk}} = P_d^{\text{fix}} + \gamma_+ [\tilde{p}_{dht}^{\text{risk}} - P_d^{\text{fix}}]_+$, with $\gamma_+ = E_{\text{act}}(\mathcal{W}_{\text{lock}}) / \sum_{d,h,t} [\tilde{p}_{dht}^{\text{risk}} - P_d^{\text{fix}}]_+ \Delta t$. For locked day h , region d , and interval t , define the fixed linear dual-price surrogate $\pi_{dht} = r^{\text{DR}} [\tilde{p}_{dht}^{\text{risk}} - P_d^{\text{fix}}]_+ / \tilde{\Pi}_{\text{max}}$, where $\tilde{\Pi}_{\text{max}} = \max_{d,h,t} [\tilde{p}_{dht}^{\text{risk}} - P_d^{\text{fix}}]_+$. Re-enumerating every declaration-feasible block with the waiting, event-tariff, and π_{dht} terms produces the exact minimizer of this finite linear-price objective. The optimality statement concerns the declaration-feasible policy; equality to the aggregate target is a separate primal residual. The bridge therefore records, for every locked day,

$$\begin{aligned} e_{dht}^{\text{br}} &= p_{dht}^{\text{exec}} - \tilde{p}_{dht}^{\text{risk}}, \\ \text{nRMSE}_{\text{flex}} &= \frac{\|e^{\text{br}}\|_2}{\max\{\|\tilde{p}^{\text{risk}} - P^{\text{fix}}\|_2, \epsilon\}}. \end{aligned} \quad (8)$$

along with the total-load normalization and the flexible relative-L2 denominator. An independent enumeration checks all 7,306,622 candidate starts, verifies the stored minimizer, and reports these residuals. The bridge stores raw/scaled profiles, the common digest, daily residuals, and the central-energy check; it is a deterministic realization of the fitted target under the declared surrogate, not a second aggregate optimization. For the job-to-network certificate, the indexed witness is mapped without a second optimization. To avoid overloading the deadline endpoint d_j , let $\mathcal{J}_{sk\delta}$ be the ledger partition whose source, class, and declared destination labels are s, k, δ ; the evaluated witness uses the native assignment $\delta = s_j$ and therefore has no outcome-dependent migration:

$$x_{sk\delta\tau}^{\text{job}} = \sum_{j \in \mathcal{J}_{sk\delta}: r_j \leq \tau < d_j} u_{j\tau}, \quad (9)$$

$$p_{\delta\tau}^{\text{job}} = P_{\delta}^{\text{fix}} + \Delta t^{-1} X_{\delta\tau}^{\text{job}}, \quad X_{\delta\tau}^{\text{job}} = \sum_{s,k} x_{sk\delta\tau}^{\text{job}}. \quad (10)$$

Aggregate migration remains a matched-workload control; (10) is the indexed replay profile. The audit reconstructs X^{job} from every stored $u_{j\tau}$ and verifies equality before network valuation; no separately optimized aggregate receives a network value. Workload deferral and geographic assignment follow [5], [6], while rolling-state control follows [9]; waiting and migration costs are $w_k(\tau - t)x_{skd\tau}$ and $m_{sd}x_{skd\tau}$, with service-time substitution differing only by a fixed-arrival constant.

The submit-time and post-event ledgers have separate canonical digests. Each submitted record is represented by the scheduler-only tuple; its Secure Hash Algorithm 256 (SHA-256) digest is

$$\begin{aligned} \ell_j^{\text{sub}} &= (\text{id}_j, t_j^{\text{submit}}, \text{timelimit}_j, g_j^{\text{req}}, \kappa_j, \sigma_j), \\ \mathcal{L}_{\text{sub}} &= \text{sort}\{\ell_j^{\text{sub}} : j \in \mathcal{J}_{\text{sub}}\}, \\ h_{\text{sub}} &= \text{SHA-256}(\text{serialize}(\mathcal{L}_{\text{sub}})). \end{aligned} \quad (11)$$

where g_j^{req} , κ_j , and σ_j are the requested-GPU, class, and scheduler-state fields. The resulting canonical commitment is (11). Start/end times, DCGM energy, and telemetry counts enter

a separate execution ledger whose join and energy equality are checked only after commitment. Canonical ordering makes h_{sub} independent of execution telemetry [15]. Measured power is reconciled with the 118-MW nameplate fixed before the split.

a) *Information and threat model.*: At decision time, $\mathcal{I}^{\text{dec}}$ contains the submission ledger, declared topology, pre-event prices, and configuration; the scoring set adds the closed meter and execution ledger only after the contract is frozen. The threat model covers post-commitment edits, replayed records, and attempts to introduce future arrivals into a committed response. The canonical digest rejects an altered scheduler tuple, the information-boundary certificate rejects future or locked-test fields, and the settlement rule caps payment with the closed meter. Scheduler and DCGM services remain trusted sources: a digest detects a changed field, while upstream attestation is required to establish that a submitted deadline, region, class, or energy field was semantically truthful. Complete-ledger profiles are therefore ex-post information comparators; the slot-62 committed ledger is the deployable contract protocol.

B. Counterfactual, response, and false credit

For event set $\mathcal{E} \subset \mathcal{T}$, p denotes regional power in MW and time sums use Δt (MWh). The declaration-causal estimate is $\tilde{p}$; p^0 is the declaration-only strategic reference used for mechanism isolation, $p^{1,\text{sim}}$ is the independently solved event-response schedule, and p^{obs} is the Massachusetts Institute of Technology (MIT) Data Center GPU Manager (DCGM) execution trace. The observed trace is opened only after a profile is frozen; no utility event label exists, so it is never a causal input. The mechanism-isolation response $r = p^0 - p^{1,\text{sim}}$ is signed and retains rebound. For $z \in \{\text{obs}, \text{sim}\}$, submitted and true credit are $\hat{r}^{(z)} = [\tilde{p} - p^{(z)}]_+$ and $r^{(z)} = [p^0 - p^{(z)}]_+$; false credit is

$$F^{(z)} = \Delta t \sum_{d,t \in \mathcal{E}} [\hat{r}_{dt}^{(z)} - r_{dt}^{(z)}]_+, \quad z \in \{\text{obs}, \text{sim}\}. \quad (12)$$

The simulated source defines the validation epigraph; the observed source is recomputed only in locked replay. False credit is computed from (12) in absolute MW-slots, avoiding a ratio with a zero-credit denominator. Before the event the operator freezes a contract baseline p^{con} . After meter closure, define the submitted gross reduction, the contract-cap reduction, and the payable reduction by

$$\begin{aligned} q_{dt}^{\text{gross}} &= [\tilde{p}_{dt} - p_{dt}^{\text{obs}}]_+, \\ q_{dt}^{\text{cap}} &= [p_{dt}^{\text{con}} - p_{dt}^{\text{obs}}]_+, \\ q_{dt}^{\text{pay}} &= \min\{q_{dt}^{\text{gross}}, q_{dt}^{\text{cap}}\}. \end{aligned} \quad (13)$$

The payable response energy is $Q_{\text{MWh}}^{\text{pay}} = \Delta t \sum_{d,t \in \mathcal{E}} q_{dt}^{\text{pay}}$, and its capacity cash component is $G^{\text{cap}} = r^{\text{DR}} Q_{\text{MWh}}^{\text{pay}}$. Equation (13) uses only the submitted baseline, frozen contract, and closed meter. Here $p^{\text{con}} = p^{\text{safe}}$ for the payment-band panel and is the no-tariff committed-ledger profile for the decision-time panel; p^0 remains an offline diagnostic. On locked event slots, $\text{nRMSE} = \sqrt{N_E^{-1} \sum_n (\tilde{p}_n - p_n^0)^2} / \max\{N_E^{-1} \sum_n p_n^0, \epsilon\}$. With $M = \sum_n \min\{\hat{r}_n, r_n\}$, precision, recall, and F_1 are $M / \max\{\sum_n \hat{r}_n, \epsilon\}$, $M / \max\{\sum_n r_n, \epsilon\}$, and $2PR / \max\{P + R, \epsilon\}$ over the N_E region-slot samples.

The reported error metrics use mean absolute error (MAE) and root-mean-square error (RMSE); normalized RMSE (nRMSE) is the scale-free form above. The risk simplex is the sole convex quadratic program (QP); all released workload counterfactuals and payment caps use the linear programs stated above.

C. Strategic reference scheduling

Consider an arithmetic mean of $H = 10$ reference days, event probability q , and DR price π^{DR} . If reference day j adds event-window service δR_j , the expected payment change is

$$\Delta \Pi = \frac{q\pi^{\text{DR}}}{H} \sum_{j=1}^H \delta R_j - \sum_{j=1}^H \Delta C_j. \quad (14)$$

Let c'_j be the right derivative of minimum operating cost with respect to the event-window service constraint on day j . Linear-program sensitivity [16] gives

$$\frac{q\pi^{\text{DR}}}{H} > \min_j c'_j \iff \exists \text{ a strictly profitable local deviation.} \quad (15)$$

At a degenerate optimum the current dual can be zero while the right marginal cost is positive; an independent check evaluates c'_j with 5×10^{-4} and 10^{-3} MWh finite differences (agreement tolerance 10^{-6} USD/MWh). Equations (14)–(15) specialize the historical-baseline premise in [4].

D. Network value

Let M be the data-center-to-bus incidence matrix and $p_t = p_t^{\text{native}} + Mp_t^{\text{dc}}$ the nodal demand vector. For generator-to-bus map C_g , generator bounds, and power-transfer distribution factor (PTDF) H with line ratings $\bar{f}$, the piecewise-linear security-constrained economic dispatch (SCED) value is

$$\begin{aligned} V(p) = \min_g \sum_i C_i(g_i) \\ \text{s.t. } \mathbf{1}^\top g = \mathbf{1}^\top p, \quad g \leq \bar{g}, \\ -\bar{f} \leq H(C_g g - p) \leq \bar{f}. \end{aligned} \quad (16)$$

The incidence mapping aggregates demand at its connection bus; (16) is the standard lossless DC representation [11], [12]. The realized signed grid value is

$$\Delta V = V(p^0) - V(p^1). \quad (17)$$

Positive ΔV is a credit and negative ΔV a debit; no positive-only truncation is applied.

III. PROPOSED METHODOLOGY

A. Model-based framework and matched-information pre-estimation

The state, power, and network equations define the feasible state before a payment target is formed (Fig. 1). Meter history and submitted declarations enter the estimator in parallel; execution remains scoring-only. Lagged meter/calendar baselines and declaration-view ledger comparators include ridge, gradient boosting, and extremely randomized trees [17], [18]. Synthetic control and nonnegative donor weighting are evaluated as a separate history-based class [19], [20]. The quantile output receives the same exact projection, with its penalty selected by contiguous folds.

B. Sparse cumulative workload-state verifier

For statistical estimate $\tilde{p}$ and each predeclared penalty ρ_ℓ , the first-stage program solves

$$\min_{x,y,z} C_w(x) + \rho_\ell \sum_{d,t} z_{dt} \quad \text{s.t.} \quad z_{dt} \geq \pm(p_{dt}(x) - \tilde{p}_{dt}), \quad (18)$$

subject to (1)–(4). The absolute-value epigraph is standard [16]; its use for workload-ledger verification is specific here. C_w is workload cost, C_i is the generator-segment cost in (16), and $V(p)$ is dispatch value, so (18) changes only the objective. Six globally solved metadata schedules $p^{(\ell)}$ share identical arrivals and constraints. A seventh causal Pareto projection is selected on validation from the high-5-of-10 declaration target by a lexicographic rule over nRMSE, false credit, under-credit, and $1 - F_1$. The risk simplex contains these seven causal profiles. A complete-ledger feasible-quantile projection is retained as an external post-commitment comparator and is used only in the separate payment hull. For a hull profile, $\bar{p} = \sum_{\ell=1}^L \alpha_\ell p^{(\ell)}$, $\alpha \geq 0$, $\mathbf{1}^\top \alpha = 1$, remains feasible because the common scheduling set is convex.

The convex fit minimizes validation error plus total and daily-tail false-credit exposure, measured by conditional value at risk (CVaR) [21]. Let ℓ^* minimize contiguous-fold nRMSE and let B_h be the absolute daily false-credit exposure of that single feasible projection on validation day h . For a candidate simplex profile define $F_h(\alpha) = \sum_{n \in h} f_n$, where f_n is the positive-part false credit at sample n . The risk fit uses absolute MW-slot exposure throughout; there is no ratio with a potentially zero true-credit denominator. With $m = \lceil (1 - \gamma)H \rceil$ and reserve $\beta \in (0, 1]$, solve

$$\begin{aligned} \min_{\alpha, f, \nu, u} \quad & \frac{\|\bar{p} - p^{0, \text{decl}}\|_2^2}{N s_p^2} + \lambda \|\alpha\|_2^2 + \omega_T \frac{\sum_{h=1}^H F_h}{S_T} \\ & + \omega_C \frac{\nu + m^{-1} \sum_{h=1}^H u_h}{S_C}, \end{aligned} \quad (19)$$

Equation (19) fixes both preferences inside the fit. Here $s_p^2 = \max\{N^{-1} \|p^{0, \text{decl}}\|_2^2, \epsilon\}$, $S_T = \max\{\sum_h B_h, \epsilon\}$, and $S_C = \max\{m^{-1} \sum_{h \in \text{tail}_\gamma} B_h, \epsilon\}$. Single-axis ablations remove one term and its constraint; the locked joint fit uses $\omega_T = 1.00$, $\omega_C = 0.50$, $\gamma = 0.75$, and $\lambda = 10^{-8}$. The total-exposure term aggregates all event slots, while the CVaR term controls the daily tail; both weights are fixed before locked scoring. The budgets are

$$\sum_{h=1}^H F_h(\alpha) \leq \beta \sum_{h=1}^H B_h, \quad (20)$$

$$\text{CVaR}_{0.75}(F_h(\alpha)) \leq \beta \text{CVaR}_{0.75}(B_h). \quad (21)$$

The positive part is represented without differentiating through a sort. For each event sample n and validation day h , introduce $f_n \geq 0$, a CVaR threshold $\nu \geq 0$, and tail slacks $u_h \geq 0$:

$$f_n \geq p_n^\alpha - p_n^{(\text{sim})} - r_n^{(\text{sim})}, \quad (22)$$

$$F_h(\alpha) = \sum_{n \in h} f_n, \quad (23)$$

$$F_h(\alpha) - \nu - u_h \leq 0, \quad (24)$$

$$\nu + \frac{1}{m} \sum_h u_h \leq \beta \text{CVaR}_{0.75}(B_h). \quad (25)$$

The linear CVaR epigraph in (25) uses the same m -day denominator as the empirical statistic and avoids sorting. Here

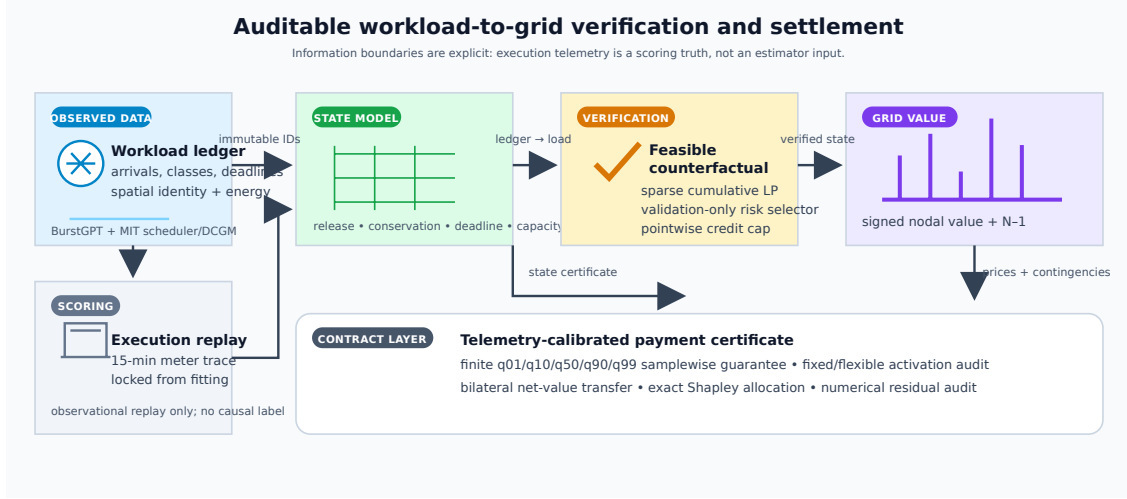


Fig. 1. Verification-to-settlement framework. Validation labels select the counterfactual risk contract; locked execution truth is used only for scoring. The right branch computes aggregate N-1 value and exact participant shares.

$r_n^{(\text{sim})} = [p_n^0 - p_n^{(\text{sim})}]_+;$ the implementation uses the equivalent threshold $p_n^{1,\text{sim}} + r_n^{(\text{sim})}$ and stores the identity audit. To preserve the point-estimate metric, let $q_\ell = N^{-1/2} \|p^{(\ell)} - p^0, \text{decl}\|_2$ and $q_\star$ denote the RMS error of the validation-selected causal Pareto reference. The simplex additionally satisfies the linear sufficient condition

$$\sum_{\ell=1}^L \alpha_\ell q_\ell \leq \sqrt{1 + \varepsilon_{\text{acc}}} q_\star, \quad \varepsilon_{\text{acc}} = 0.10. \quad (26)$$

By the triangle inequality, (26) bounds the actual simplex RMS error by $\sqrt{1 + \varepsilon_{\text{acc}}} q_\star$, equivalent to a 10% validation-MSE budget. It therefore prevents a risk improvement from being purchased with an unreported forecast degradation while retaining a non-degenerate feasible simplex. The full problem is solved by the declared CVXPY/Clarabel convex-QP path. Independent Karush–Kuhn–Tucker stationarity, epigraph, simplex, accuracy, and primal residuals are recomputed after the solve. Four contiguous folds select β only when each training contract is feasible. A separately reported payment-contract projection targets $\bar{p}$ while imposing

$$p_{dt}^{\text{safe}} \leq p_{dt}^{(\ell^*)}. \quad (27)$$

The pointwise cap in (27) is the upper side of the contractual credit band; it is evaluated on every event slot before settlement and is not substituted for the scientific risk profile.

a) *Two-sided credit band.*: To exclude a zero-credit envelope, tie a lower band to the fitted target. Define $p_{dt}^{\text{floor}} = \min\{p_{dt}^{(\ell^*)}, \bar{p}_{dt}\}$, where $\bar{p}$ is the pre-projection validation fit. The event-slot constraints are

$$p_{dt}^{\text{floor}} - \varepsilon \leq p_{dt}^{\text{safe}} \leq p_{dt}^{(\ell^*)}, \quad (d, t) \in \mathcal{D} \times \mathcal{E}, \quad (28)$$

The release sets $\varepsilon = 0$; the lower side is therefore the exact pointwise minimum of the two independently formed profiles, and the selected single projection proves nonemptiness. Equations (20)–(21) control aggregate and daily-tail exposure, while the lower side in (28) regularizes under-credit and gives no unseen-trace guarantee. Realized overpayment is handled by (13).

b) *Decision-time information boundary.*: A gate removes all post-gate arrivals before refitting the matched-information

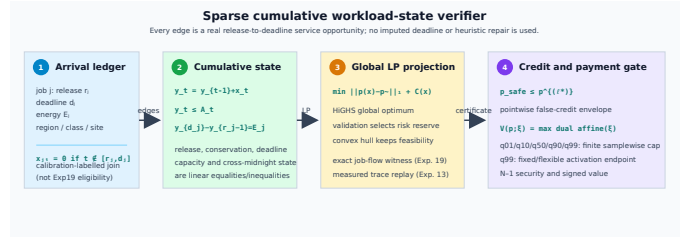


Fig. 2. Sparse cumulative verifier detail. Each admissible edge is a declared release-to-deadline service opportunity; no imputed deadline or heuristic repair is used. Every edge is a real release-to-deadline service opportunity; no imputed deadline or heuristic repair is used.

target. The locked protocol uses slot 62, 30 minutes before the first event slot, and a no-tariff LP freezes the contract baseline with a predeclared 3-MWh minimum of participating event-window service. A second linear program (LP) applies the selected positive DR price to the same masked ledger while carrying unexpired state. Reserved future jobs cannot be credited, and the tariff plan is scored against the closed meter through (13). The independent execution trace is read only in the scoring pass.

To separate capacity planning from payment eligibility, the gate publishes a causal reserve envelope. For candidate quantile η and event day d ,

$$\hat{a}_{sk\tau}^{\text{res}}(d; \eta) = Q_\eta(\{a_{sk\tau}(h) : h < d\}), \quad \tau \geq t_{\text{gate}}, \quad (29)$$

Equation (29) uses earlier days only. The predeclared candidate set is $\{0.55, 0.60, 0.65, 0.70\}$ with a 1.0-MWh/day validation false-credit allowance; it selects $\eta = 0.55$ before locked scoring. The reserve is a planning envelope, and no reserved job enters payment before commitment.

The cumulative state has $O(|\mathcal{S}||\mathcal{K}||\mathcal{D}||\mathcal{T}|)$ nonzeros, versus quadratic explicit prefixes. Equation (5) retains every observed future arrival for the 512-slot deadline. Two lexicographic LPs (minimum L1, then operating cost) embed the day trajectory and are solved globally with HiGHS [22]. The resulting state-to-credit path is shown in Figs. 1 and 2; execution and interval audits are separate (Experiments 13–15).

C. Signed nodal and N-1 settlement

Let $\lambda^0 \in \partial V(p^0)$ be the nodal demand-value subgradient. Signed linear settlement is $(\lambda^0)^\top (p^0 - p^1)$, while (17) is the realized signed system-value benchmark. Convexity [16] gives

$$0 \leq (\lambda^0)^\top (p^0 - p^1) - \Delta V = V(p^1) - V(p^0) - (\lambda^0)^\top (p^1 - p^0). \quad (30)$$

Thus the linear rule upper-bounds exact value in the polyhedral market model; equality requires a common supporting subgradient.

For market-compatible remuneration, let $\mathcal{W}$ contain the complete response cycle from anticipatory shift through recovery. Its space-time remuneration is

$$\Pi^{\text{ST}} = \Delta t \sum_{t \in \mathcal{W}} \sum_d \lambda_{dt}^0 (p_{dt}^0 - p_{dt}^1). \quad (31)$$

This is the nodal virtual-link structure of [7], [8]; summing site payments reproduces (31).

Space-time energy remuneration is one component of the contracted DR product. Let $\mathcal{D}^{\text{part}}$ be the predeclared participating sites, $\mathcal{E}$ the event intervals, and r^{DR} the cleared service price. The signed delivered capacity service and total operator value are

$$Q^{\text{cap}} = \Delta t \sum_{d \in \mathcal{D}^{\text{part}}} \sum_{t \in \mathcal{E}} (p_{dt}^0 - p_{dt}^1), \quad (32)$$

$$G = r^{\text{DR}} Q^{\text{cap}} + \Pi^{\text{ST}}, \quad (33)$$

where p^0 is the minimum-cost no-event schedule on the same window. Equation (32) defines the bilateral capacity product and need not equal trace-anchored p^0 ; Π^{ST} debits spatial transfer and recovery. This signed service model follows demand-response mechanism design [23], [24].

Let x^1 and x^0 induce p^1 and p^0 . Define the participant opportunity cost $C_{\text{opp}} = C_w(x^1) - C_w(x^0)$ and surplus $S = G - C_{\text{opp}}$. With zero disagreement utilities, the symmetric Nash contract [25] is

$$z = \mathbb{I}\{S \geq 0\}, \quad P^{\text{B}} = z \left(C_{\text{opp}} + \frac{S}{2} \right), \quad u_{\text{dc}} = u_{\text{op}} = z \frac{S}{2}. \quad (34)$$

Equation (33) combines the capacity payment and signed space-time value. Equation (34) uses nonactivation as the outside option, keeps both utilities nonnegative, and is budget balanced. It is separate from independent system operator (ISO) nodal remuneration and the exact dispatch-value benchmark. For the executable full-day ledger, each interval first retains the signed contract value $s_{dt} = \Delta t r^{\text{DR}} q_{dt}^{\text{pay}} + \Delta V_{dt}$, where q_{dt}^{pay} is in MW, r^{DR} is in USD/MWh, Δt is in hours, and both terms are in USD. The reported economic ledger is $S^{\text{signed}} = \sum_{d,t} s_{dt}$; a cash presentation $S^{\text{cash}} = \sum_{d,t} \max\{s_{dt}, 0\}$ is stored separately. The cash floor therefore leaves negative network-value debits visible in the signed cycle total. The complete-cycle ledger $\mathcal{W}$ includes pre-event increases and rebound, so workload conservation gives $\sum_{d,t \in \mathcal{W}} (p_{dt}^0 - p_{dt}^1) \Delta t = 0$ within solver tolerance.

Before payment comparison, the submitted, frozen-contract, and closed-meter reductions are intersected as follows:

$$q_{dt}^{\text{pay}} = \min\{q_{dt}^{\text{obs}}, q_{dt}^{\text{cap}}\}, \quad G = r^{\text{DR}} \Delta t \sum_{d,t \in \mathcal{E}} q_{dt}^{\text{pay}} + \Pi^{\text{ST}}. \quad (35)$$

Equation (35) is evaluated after the submitted profile, contract, and closed meter are fixed; the second line uses q^{pay} in the corresponding ledger. The observed-meter audit stores the payment chain and its budget residual (below 10^{-8} USD). The job-indexed audit stores the declaration replay $q_{dt}^{\text{replay}} = \min\{q_{dt}^{\text{dec}}, q_{dt}^{\text{cap}}\}$ to certify feasible network value. Exp30 records this role as declaration-only, and the replay's signed/cash totals remain outside payable settlement until q^{obs} is available.

Security-aware settlement augments (16) with every finite non-islanding single-line outage. If $L_{\ell k}$ is the LODF from outage k to monitored line ℓ , then

$$-\bar{f}_\ell \leq (H_\ell + L_{\ell k} H_k)(C_g g - p) \leq \bar{f}_\ell, \quad \ell \neq k. \quad (36)$$

Equation (36) is the standard preventive N-1 LODF formulation [13]; every credible row uses native ratings. Islanding rows are excluded because a connected single-reference PTDF cannot represent separate islands, consistent with the DC/N-1 planning scope [26].

D. Scenario-robust N-1 payment certificate

The pointwise envelope controls energy but not network payment because spatial loads can activate different security constraints. Let $\mathcal{P} = \text{conv}\{p^{(1)}, \dots, p^{(L)}\}$ be the feasible-profile hull and $p^{\text{cap}} \in \mathcal{P}$ the validation-selected projection. Its penalty is separate from the payment-target penalty; the feasible-quantile projection is an external comparator. Let Ξ_5 be the 1st, 10th, 50th, 90th, and 99th calibration-validation GPU-energy ratios from independent telemetry [2]. For facility profile p_d^{fac} , fixed demand P_d^{fix} , scale s_{dc} , and factor ξ ,

$$p_{dt}^{\text{dc}, \xi} = s_{\text{dc}} (P_d^{\text{fix}} + \xi [p_{dt}^{\text{fac}} - P_d^{\text{fix}}]), \quad p_{dt}^{\text{fac}} \geq P_d^{\text{fix}}. \quad (37)$$

The scale uses the calibration-validation q_{99} flexible peak, keeping the fixed component invariant to ξ ; raw endpoints remain activation scenarios. The injection is $M p_t^{\text{dc}, \xi}$, with superscripts α and "cap" applying (37) to p^α and p^{cap} . For each settlement day, the payment-certified verifier defines $B_\xi^{\text{cap}} =$

$$\begin{aligned}
& \Delta t \sum_{t \in \mathcal{E}} V_{N-1}(p_t^{\text{native}} + M p_t^{\text{dc}, \xi, \text{cap}}) \text{ and solves} \\
& \text{lexmin}_{\alpha, e, \eta, \{g_t^\xi\}} \left(\sum_{d, t \in \mathcal{E}} e_{dt}, -\eta \right) \quad (38) \\
& \text{s.t. } p^\alpha = \sum_{\ell=1}^L \alpha_\ell p^{(\ell)}, \\
& \alpha \geq 0, \quad \mathbf{1}^\top \alpha = 1, \quad 0 \leq \eta \leq 1, \\
& e_{dt} \geq \pm(p_{dt}^\alpha - p_{dt}^{\text{safe}}), \\
& g_t^\xi \text{ satisfies (16) and (36),} \\
& \quad \text{with } p_t = p_t^{\text{native}} + M p_t^{\text{dc}, \xi, \alpha}, \\
& \quad \quad t \in \mathcal{E}, \quad \xi \in \Xi_5, \\
& \Delta t \sum_{t \in \mathcal{E}} \sum_i C_i(g_{it}^\xi) \\
& \quad \leq (1 - \eta) B_\xi^{\text{cap}}, \quad \xi \in \Xi_5.
\end{aligned}$$

The piecewise-linear epigraph is solved by two global LPs: trajectory error is minimized first, then the common fractional margin η . Feasibility is the N-1 upper bound and p^{cap} proves nonemptiness. Hence, for any $\xi \in \Xi_5$ and realized load $p^{1, \xi}$, daily payment obeys

$$\begin{aligned}
& \sum_{t \in \mathcal{E}} \Delta t [V_{N-1}(p_t^{\text{native}} + M p_t^{\text{dc}, \xi, \alpha}) - V_{N-1}(p_t^{1, \xi})] \\
& \leq \sum_{t \in \mathcal{E}} \Delta t [V_{N-1}(p_t^{\text{native}} + M p_t^{\text{dc}, \xi, \text{cap}}) - V_{N-1}(p_t^{1, \xi})]. \quad (39)
\end{aligned}$$

The realized cost cancels, so this is a samplewise certificate without execution truth. Every finite non-islanding contingency is imposed using standard SCED/LP primitives [12], [16]. The hull contains six first-stage projections and the matched feasible-quantile comparator. risk-constrained verifier is the validation-fitted simplex point under the two risk budgets; the separately stored payment contract targets it under the pointwise cap and is excluded from risk fitting.

The finite cap also has a set-valued certificate. Let $p^{\text{cap}} = p^{(\ell^*)}$, p^{pay} be the separately selected payment target, and $p^{\text{cert}} = p^\alpha$. For $\Delta V_\xi(p) = V_{N-1}(p, \xi) - V_{N-1}(p^{1, \xi})$ and $p_\theta = (1 - \theta)p^{\text{cap}} + \theta p^{\text{cert}}$, the interval is

$$\mathcal{I}_\xi = \left[\min_{0 \leq \theta \leq 1} \Delta V_\xi(p_\theta), \max\{\Delta V_\xi(p^{\text{cap}}), \Delta V_\xi(p^{\text{cert}})\} \right]. \quad (40)$$

The lower endpoint in (40) is one joint N-1 SCED LP with a shared segment parameter; convexity gives the endpoint maximum and the cap belongs to $\mathcal{I}_\xi$. Validation-frozen profiles and the realized counterfactual determine both endpoints; the interval width is uncertainty cost, while (38) remains the finite-scenario certificate.

E. Participant allocation

For participants N , let $v(S)$ be signed dispatch value when coalition S moves from baseline to actual demand. The exact allocation is

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [v(S \cup \{i\}) - v(S)]. \quad (41)$$

This Shapley specialization in (41) uses signed grid value [27]; coalition-size grouping gives $\sum_i \phi_i = v(N)$ when

$v(\emptyset) = 0$. All 16 four-site and 256 eight-portfolio coalitions are enumerated exactly. For larger contracts, exchangeable site slices retain binomial multiplicities and reduce evaluation to $\prod_g (m_g + 1)$ count states (tested through 20 participants).

F. Analytical guarantees

These certificates delimit the logical scope. Under the interior and differentiability assumptions of (14), the threshold in (15) is exact, as verified on the full 56-cell grid. The workload set, simplex schedules, and two-sided envelope are feasible. For any observed execution c , define credit from baseline p as $r(p; c) = [p - c]_+$ and the oracle baseline as p^0 . The pointwise envelope in (28) gives

$$\begin{aligned}
& [r(p_{dt}^{\text{safe}}; c_{dt}) - r(p_{dt}^0; c_{dt})]_+ \leq [r(p_{dt}^{(\ell^*)}; c_{dt}) - r(p_{dt}^0; c_{dt})]_+, \\
& [r(p_{dt}^0; c_{dt}) - r(p_{dt}^{\text{safe}}; c_{dt})]_+ \leq [p_{dt}^0 - p_{dt}^{\text{floor}}]_+ + \varepsilon.
\end{aligned}$$

The first inequality bounds excess credited response. For the second, $r(\cdot; c)$ is monotone and 1-Lipschitz, so the lower side of (28) bounds missed credit by the oracle-to-floor gap and the declared band tolerance. Summation gives the corresponding ledger bounds without locked outcomes.

Base-case and stacked N-1 values are convex piecewise-affine functions of nodal demand; (30) gives the linear-value bound. Each dispatch embedded in (38) is feasible, so (39) holds for the five frozen conversion scenarios Ξ_5 . In the continuous segment audit, the realized counterfactual is fixed while the affine segment between the two frozen contract profiles varies; a joint LP gives the exact segment minimum, and convexity places its maximum at an endpoint. No endpoint assertion is made when conversion simultaneously scales the realized counterfactual. The payment-hull and segment audits use independent ten- and four-segment models.

The digest detects post-commit scheduler-tuple changes, while the execution join audits ordering and energy conservation. An optimum of (6) gives declaration-only runtime, capacity, and nameplate feasibility without aggregate rounding. With $B_{dj} = \mathbf{1}\{s_j = d\}$ and bus map M , the typed residuals are

$$\begin{aligned}
r_j^{\text{job}} &= \sum_{\tau=r_j}^{d_j-1} u_{j\tau} - E_j^{\text{run}}, \\
r_{d\tau}^{\text{agg}} &= X_{d\tau} - \sum_j B_{dj} u_{j\tau}, \\
r_{b\tau}^{\text{map}} &= p_{b\tau} - P_{b\tau}^{\text{fix}} - \Delta t^{-1} \sum_d M_{bd} X_{d\tau}.
\end{aligned}$$

The first residual uses the same E_j^{run} as (6); $\hat{E}_j$ remains diagnostic. Residuals at most ε identify the witness before N-1 valuation; allocation checks $\sum_i \phi_i = v(N)$ separately. Moreover, if the feasible SCED value is L_V -Lipschitz on the committed polytope, the mapped-value error is bounded by

$$|V(p^{\text{map}}) - V(p^{\text{job}})| \leq L_V \|r^{\text{map}}\|_1. \quad (42)$$

Equation (42) makes the zero-residual certificate explicit: the network solver receives the same nodal profile implied by the indexed jobs, rather than an independently refitted aggregate trace.

IV. EXPERIMENTAL DESIGN

A. Data, preprocessing, and information boundary

BurstGPT provides 5,188,507 successful requests and token service [1]; MIT Supercloud provides scheduler submissions and independent Data Center GPU Manager (DCGM) energy [2]. The population contains 75,326 valid submissions and 71,128 positive-energy execution matches; 68,662 have valid intervals in the common 121-day tensor. Submission time defines arrivals, while measured intervals are scoring truth. The GPU-power model uses 44,391 training, 9,637 calibration-validation, and 41,154 locked jobs. Calibration-validation ratios define the five scenarios Ξ_5 in (38); execution joins are opened only after the declaration witness is frozen. The 13 positive scheduler/DCGM pairs without a valid interval are calibration labels only and cannot enter replay or selection. The declared runtime plus a precommitted 96-slot queue allowance defines the contract window; population membership is not filtered by outcomes.

The traces do not identify facility geography. Four-region coupling is therefore a trace-driven benchmark with feature-stratified round-robin assignment. We enumerate all 24 region-to-bus permutations and two concentration controls at three penetrations. The Power Grid Library (PGLib) native-load multiplier is fixed at 0.75 by the intact and contingency feasibility rule; all scored cells are feasible. A separate fixed-generator-bus panel uses the pre-registered 0.90 multiplier and remains conditional on that scenario. Payment-network scale is calibrated on q_{99} at 4% peak penetration, and all raw endpoints are N-1-feasible under this scale. The flexible-only capacity-safe factor is reported as a planning diagnostic, not an endpoint clip. PGLib and MATPOWER supply case data from their public benchmark libraries [12], [28]. The IEEE Reliability Test System (RTS-24) supplies the third network case [29].

B. Baselines, parameters, and inference

The 54-day comparison includes meter and statistical baselines, feasible quantile projections, synthetic control, the selected single projection, and the joint and single-axis risk fits. Baselines share the declaration ledger; execution truth is withheld until scoring. Deferral windows are 0, 4, and 512 slots, with 118-MW flexible nameplate and a 6-MW fixed component per region. Projection penalties and normalized risk preferences are selected on contiguous validation folds. The joint profile must satisfy a predeclared 10% MSE non-inferiority bound and total/CVaR exposure budgets. A separate slot-62 gate commits 3 MWh. Payment uses an independent ten-segment N-1 evaluator; the contractual certificate uses four segments. F_1 is the harmonic mean of precision and recall. 5,000-replication moving three-day bootstrap intervals and all 2^{18} block-sign assignments use Holm correction [30], [31].

Sparse-system size ranges from 5,760 variables and 13,424 nonzeros at 96 slots to 36,480 and 85,488 at 608 slots, with solve times of 0.039–0.212 s. Release runs clamp each numerical backend to one thread; worker pools use at most 12 threads, below the 20-core limit. Long panels checkpoint schema- and profile-validated results. The AC model enforces

active/reactive balance, voltage and reactive-generation bounds, and apparent-power limits [12]. For each outage, non-reference active generation is fixed to its intact value while the reference generator, reactive outputs, and voltages remain recourse; all feasible cells pass the stated voltage, thermal, and fixed-plan checks.

The evaluation spans strategic-baseline incentives, signed nodal settlement, N-1 and AC feasibility, independent meter replay, declaration-indexed job scheduling, and provenance checks. All comparisons use the same locked days unless a deterministic network or ledger rule applies. An independent audit reconstructs the job-to-region aggregation, verifies common-witness digests and source hashes, and checks that risk, payment, and outage artifacts retain their separate information roles.

V. RESULTS

A. Incentive boundary and independent verification

The strategic-baseline test solves the entire 8×7 DR-price/call-probability grid. The dual threshold (15) agrees with the independently solved strategic program in all 56 cells. At \$150/MWh and $q = 0.35$, strategic reference scheduling inflates the event baseline by 8.223 MWh. Of 55.452 MWh credited, 44.458 MWh is unsupported by physical reduction, giving a false-credit ratio of 0.802. The profitable region follows the derived marginal boundary and does not use a fitted classifier; the complete grid and its phase diagram are retained in the experiment output.

Twelve external and internal reference methods are evaluated on the locked 54-day mechanism-isolation panel (Table I). Relative to the metadata projection at $\rho = 10$, the joint verifier reduces nRMSE, false credit, and under-credit and raises F_1 . That row is not the single profile used to anchor the risk fit: the validation-selected Causal Pareto reference has locked nRMSE/false credit/under-credit/ F_1 of 0.6914/1.895/17.024/0.699. Against this internal reference, the joint fit lowers false credit but has higher nRMSE and under-credit and lower F_1 . The total-budget-only ablation (0.8917/0.6455/26.3723/0.5490) improves all four reported metrics over the joint fit and also has lower false-credit CVaR (2.222 versus 2.607 MWh/day). Thus the locked panel does not establish that the tail constraint adds predictive or empirical tail-risk benefit. The joint profile remains the numerical optimizer of the declared seven-profile convex program and meets its two validation-frozen budgets; those facts do not imply dominance over the ablations on locked outcomes. The validation certificate records reference/fitted MSE of 188.932/200.575 against the frozen 207.825 bound, KKT residual 5.80×10^{-10} , primal residual 3.74×10^{-11} , and satisfied exposure budgets. Complete-ledger rows are post-event diagnostics; they do not enter fitting or reserve selection. On the 54-day paired three-day block bootstrap (5,000 draws), joint minus total-only mean false credit is +0.134 MWh/day (95% interval 0.016–0.287); the 75%-CVaR difference is +0.394 MWh/day (95% interval -0.049–0.923). The first interval excludes zero and the second includes it. The locked ablation therefore gives no evidence that the CVaR term reduces false credit or tail risk.

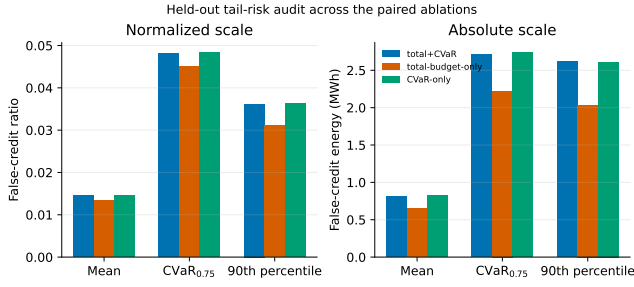


Fig. 3. Result evidence for the two coupled certificates. Left: held-out normalized and absolute false-credit tails for the joint, total-only, and CVaR-only fits. Right: the locked-day aggregate contract target and the declaration-indexed executable realization, with its audited residual below; the realization is the exact finite-start policy minimizer, while target tracking is audited rather than assumed.

TABLE I

LOCKED 54-DAY MECHANISM-ISOLATION RESPONSE RESULTS (DAILY MEANS). EXTERNAL STATISTICAL COMPARATORS AND INTERNAL RISK REFERENCES ARE SHOWN TOGETHER; EX-POST ROWS ARE INFORMATION-BOUNDARY DIAGNOSTICS.

Method	nRMSE	False MWh	Under-credit	F_1
High-5-of-10	1.586	41.220	11.169	0.572
Ridge	1.038	21.906	25.226	0.545
Gradient Boosting	1.011	20.977	26.401	0.534
Extra Trees	1.090	26.629	18.176	0.597
Metadata Gradient Boosting	0.774	9.183	21.528	0.640
Ex-post Metadata Gradient Boosting	0.750	9.171	20.817	0.643
Ex-post Quantile Gradient Boosting	1.337	0.591	54.391	0.237
Synthetic Control	0.782	9.394	21.678	0.676
Feasible Quantile Projection	1.344	0.098	52.647	0.225
Single Feasible Projection	0.943	2.470	26.812	0.463
Causal Pareto risk reference	0.6914	1.895	17.024	0.699
Total-budget-only verifier	0.8917	0.646	26.372	0.549
Tail-Risk Feasible Counterfactual	0.9051	0.824	26.548	0.532
Risk-Constrained Convex Verifier	0.9021	0.780	26.506	0.536

TABLE II

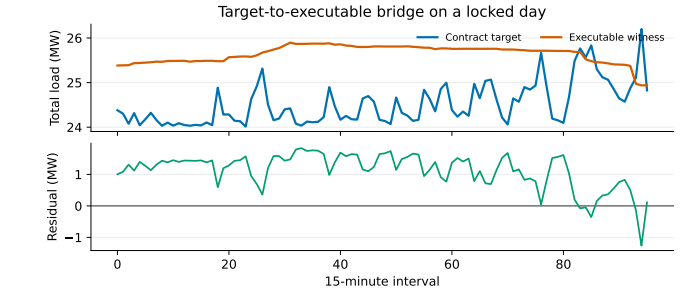
BASE-CASE AND COMPLETE N-1 SETTLEMENT ERRORS (USD/DAY).

Panel	Settlement	Trace MAE	Workload MAE	Trace median/overpay	Workload median/overpay
Base case	Nodal gross	360.580	421.677	—	—
Base case	Nodal exact net value	151.719	455.865	—	—
N-1	Base exact	139.221	167.367	79.79/130.42	121.01/107.69
N-1	N-1 linear	2.208	150.461	0.60/1.91	113.43/20.38
N-1	N-1 exact	0.990	150.955	0.87/0.01	113.43/20.18

The independent trace-meter replay yields event-window MAE of 1.909/6.880/7.038 MW for the feasible-quantile comparator/risk-constrained verifier/single projection over the same 54 days, separate from mechanism-isolation scoring. On the identical ledger, rolling-horizon, batch-flexibility, and coupled-service translations obtain 0.943/2.470/0.463, 1.357/0/0.183, and 1.376/0/0.111 (nRMSE/false-credit/ F_1), respectively; these are structural translations, not software reproductions [5], [10].

B. Nodal value and spatial response

The settlement analysis crosses thirteen counterfactuals with five settlement mechanisms; Table II reports the trace-anchored locked rows. Exact signed nodal value has MAE \$151.719 versus \$360.580 for nodal gross response, with workload errors \$455.865 and \$421.677. Signed netting reduces trace error by \$208.190/day (Holm-adjusted $p = 3.89 \times 10^{-4}$); uniform-to-nodal pricing changes it by $-\$1.463$ ($p = 0.0118$), and linear-to-exact value by $\$0.671$ ($p = 0.8026$). All 5,616 interval/counterfactual instances satisfy (30), with maximum residual 2.03×10^{-11} dollars.



The predeclared largest gross-minus-net offset (day 87) exposes the rebound omitted by positive-only local accounting.

C. Cross-network and N-1 security evidence

Across 6,480 four-network outcomes, exact value improves trace error in all 12 cells (mean \$1.913/day; eight Holm-adjusted tests below 0.05), with a \$0.050/day workload change. Capacity/deadline and coalition residuals are below 5.68×10^{-14} dollars. The 37-outage RTS-24 replay enforces (36), with trace/workload MAE \$139.221/\$2.208/\$0.990 and \$167.367/\$150.461/\$150.955 and maximum loading 1.000 p.u.

The complete replay scope is summarized in Table III.

The payment certificate in (39) is tested on 54 programs with a validation-frozen seven-profile hull. Contiguous folds select $\rho = 10$ for the cap and $\rho = 3$ for the payment target; all five scaling-factor certificates violate their bounds by less than 3×10^{-11} USD. The PaymentSafe/single/feasible-quantile/RiskSafe q50/q99 MAEs are 57.357/322.032, 41.141/231.841, 55.759/313.313, and 46.979/264.299 USD/day; on two held-out interior factors, RiskSafe gives 9.345/14.042 versus PaymentSafe's 9.988/15.006. Target selection excludes these transfer rows. The independent evaluator uses ten segments; the contractual certificate uses four, both fixed before locked scoring.

The AC replay has 864 connected-base and 5,400 corrective IEEE-9/14 cells, all feasible. A fixed-plan contingency panel adds 9,612 method-domain admissible fixed-plan contingencies and 144 intact solves with zero voltage violation. The spatial-transfer panel evaluates 26 assignments at three penetrations under the pre-registered 0.75 native-load PGLib domain; risk-constrained, single, and feasible-quantile MAE triples are \$80.496/\$162.565/\$272.482, \$89.537/\$180.649/\$300.873, and \$93.084/\$187.742/\$311.952. Calibration R^2 is 0.8956/0.8196. Event-window MAE is \$26.937/\$34.166/\$8.659/\$5.325, and complete-cycle residual is \$0.621/\$0.569/\$0.104/\$0.

D. Independent trace and deployment-boundary audits

The immutable scheduler/DCGM replay contains 71,128 jobs, has a residual below 10^{-17} MWh, and carries no event tag. The slot-62 evaluation scores the complete-ledger comparator (0.902/0.780/0.536) and the committed-ledger gate (1.366/0.494/0.080). The gate masks post-gate arrivals and

TABLE III

INFORMATION-BOUNDARY AND CERTIFICATE COVERAGE. TRIPLETS ARE nRMSE/FALSE RESPONSE/ F_1 ; DECLARATION CHECKS PRECEDE LOCKED SCORING.

Panel	Cells and certificate
Complete-ledger joint fit	54 days; 0.902/0.780/0.536; mechanism-isolation scoring
Slot-62 committed ledger	54 days; 1.366/0.494/0.080; event-time profile
Indexed witness/bridge	75,326 jobs; 13,198,247 variables; 7,306,622 starts; residual $< 10^{-11}$ MWh; nRMSE 0.03023/relative-L2 1.581
DC N-1 replay	864 cells; 37 outages/cell; 1.0000 p.u.; signed/cash
AC certificates	184.664/283.470 USD IEEE-9/14: 864+5,400; four-network fixed-plan: 9,612 outages+144 intact; all admissible cells certified

execution truth before scoring and requires 3 MWh on 54 days; its score therefore measures information-boundary performance rather than payment. The independent trace-meter MAE for risk-constrained verifier is 6.880 MW. Under a separate tariff, replay precision/recall/ F_1 is 0.0210/0.0194/0.529; independent event replay reaches $F_1 = 1.000$. Table III records the evidence boundary.

E. Indexed workload counterfactual

The declaration-indexed model assigns every valid submission a runtime-complete block. Exact enumeration covers 7,306,622 feasible starts and reproduces the selected minimizer (Table III). Nameplate/central entitlements are 1,972.115/0.570 MWh, active declaration energy is 655.753 MWh, and energy residuals are below 10^{-11} MWh. Event reduction is 4.68225 MWh. The submit-time-scaled risk-constrained verifier bridge in (8) has total nRMSE 0.03023, flexible relative-L2 1.581, and maximum residual 3.220 MW. Its 5,184-cell ledger retains 1,060 debit cells, network value -0.686 USD, signed cycle 184.664 USD, and cash-floor 283.470 USD. The payment audit labels this replay declaration-only; payment still requires a closed meter.

VI. CONCLUSION

This paper couples a declaration-indexed workload verifier with signed nodal valuation. On the complete-ledger mechanism-isolation panel, the joint total-plus-CVaR fit reaches nRMSE/false credit/ F_1 of 0.902/0.780/0.536 and 26.506 MWh/day under-credit, and meets its validation-frozen exposure budgets. The locked ablation shows that the total-only fit improves all four metrics and false-credit CVaR; the joint fit also does not dominate its own Causal Pareto reference. These results leave the tail term as an unresolved design choice. Figure 3 reports the locked risk comparison and the risk-to-executable residual. At the slot-62 gate, the committed-ledger profile reaches nRMSE 1.366 and $F_1 = 0.080$, so full-ledger results do not establish event-time performance. The task-level witness covers 75,326 submissions and 5,184 RTS-24 cells; its exact-start audit reports total-load nRMSE 0.03023 but flexible relative-L2 1.581. The N-1 results are declaration-only network replays, and a closed meter is required for payment.

REFERENCES

[1] Y. Wang, Y. Chen, Z. Li, X. Kang, Y. Fang, Y. Zhou, Y. Zheng, Z. Tang, X. He, R. Guo, X. Wang, Q. Wang, A. C. Zhou, and X. Chu, "BurstGPT: A real-world workload dataset to optimize large language model serving systems," in *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, pp. 5831–5841, 2025.

[2] S. Samsi, M. L. Weiss, D. Bestor, B. Li, M. Jones, A. Reuther, *et al.*, "The mit supercloud dataset," in *2021 IEEE High Performance Extreme Computing Conference*, pp. 1–8, 2021.

[3] California Independent System Operator, "California iso baseline accuracy work group final proposal," tech. rep., California ISO, 2017.

[4] X. Wang and W. Tang, "Modeling and analysis of baseline manipulation in demand response programs," *IEEE Transactions on Smart Grid*, vol. 13, no. 2, pp. 1178–1186, 2022.

[5] Y. Cao, M. Cheng, S. Zhang, H. Mao, P. Wang, C. Li, Y. Feng, and Z. Ding, "Data-driven flexibility assessment for internet data center towards periodic batch workloads," *Applied Energy*, vol. 324, p. 119665, 2022.

[6] Y. Cao, F. Cao, Y. Wang, J. Wang, L. Wu, and Z. Ding, "Managing data center cluster as non-wire alternative: A case in balancing market," *Applied Energy*, vol. 360, p. 122769, 2024.

[7] W. Zhang, L. A. Roald, A. A. Chien, J. R. Birge, and V. M. Zavala, "Flexibility from networks of data centers: A market clearing formulation with virtual links," *Electric Power Systems Research*, vol. 189, p. 106723, 2020.

[8] W. Zhang and V. M. Zavala, "Remunerating space-time, load-shifting flexibility from data centers in electricity markets," *Applied Energy*, vol. 326, p. 119930, 2022.

[9] W. Zhang, L. A. Roald, and V. M. Zavala, "Exploring the impacts of power grid signals on data center operations using a receding-horizon scheduling model," *IEEE Transactions on Sustainable Computing*, vol. 8, no. 2, pp. 245–256, 2023.

[10] M. Chen, C. Gao, M. Shahidehpour, Z. Li, S. Chen, and D. Li, "Internet data center load modeling for demand response considering the coupling of multiple regulation methods," *IEEE Transactions on Smart Grid*, vol. 12, no. 3, pp. 2060–2076, 2021.

[11] B. Stott, J. Jardim, and O. Alsac, "DC power flow revisited," *IEEE Transactions on Power Systems*, vol. 24, no. 3, pp. 1290–1300, 2009.

[12] R. D. Zimmerman, C. E. Murillo-Sanchez, and R. J. Thomas, "MATPOWER: Steady-state operations, planning, and analysis tools for power systems research and education," *IEEE Transactions on Power Systems*, vol. 26, no. 1, pp. 12–19, 2011.

[13] D. A. Tejada-Arango, P. Sánchez-Martín, and A. Ramos, "Security constrained unit commitment using line outage distribution factors," *IEEE Transactions on Power Systems*, vol. 33, no. 1, pp. 329–337, 2018.

[14] C. Li, K. Zheng, H. Guo, C. Kang, and Q. Chen, "Computation-power coupled modeling for idcs and collaborative optimization in adns," *IEEE Transactions on Smart Grid*, vol. 15, no. 3, pp. 2762–2775, 2024.

[15] National Institute of Standards and Technology and Q. Dang, "Secure hash standard," Tech. Rep. FIPS PUB 180-4, National Institute of Standards and Technology, 2015.

[16] S. Boyd and L. Vandenberghe, *Convex Optimization*. Cambridge University Press, 2004.

[17] A. E. Hoerl and R. W. Kennard, "Ridge regression: Biased estimation for nonorthogonal problems," *Technometrics*, vol. 12, no. 1, pp. 55–67, 1970.

[18] J. H. Friedman, "Greedy function approximation: A gradient boosting machine," *The Annals of Statistics*, vol. 29, no. 5, pp. 1189–1232, 2001.

[19] P. Geurts, D. Ernst, and L. Wehenkel, "Extremely randomized trees," *Machine Learning*, vol. 63, pp. 3–42, 2006.

[20] A. Abadie, A. Diamond, and J. Hainmueller, "Synthetic control methods for comparative case studies: Estimating the effect of california's tobacco control program," *Journal of the American Statistical Association*, vol. 105, no. 490, pp. 493–505, 2010.

[21] R. T. Rockafellar and S. Uryasev, "Optimization of conditional value-at-risk," *The Journal of Risk*, vol. 2, no. 3, pp. 21–41, 2000.

[22] Q. Huangfu and J. A. J. Hall, "Parallelizing the dual revised simplex method," *Mathematical Programming Computation*, vol. 10, no. 1, pp. 119–142, 2018.

[23] B. Satchidanandan, M. Roozbehani, and M. A. Dahleh, "A two-stage mechanism for demand response markets," *IEEE Control Systems Letters*, vol. 7, pp. 49–54, 2023.

[24] M. Chen, C. Gao, M. Shahidehpour, and Z. Li, "Incentive-compatible demand response for spatially coupled internet data centers in electricity markets," *IEEE Transactions on Smart Grid*, vol. 12, no. 4, pp. 3056–3069, 2021.

[25] J. John F. Nash, "The bargaining problem," *Econometrica*, vol. 18, no. 2, pp. 155–162, 1950.

[26] North American Electric Reliability Corporation, "TPL-001-5.1: Transmission system planning performance requirements," tech. rep., North American Electric Reliability Corporation, 2020.

[27] L. S. Shapley, "A value for n-person games," in *Contributions to the Theory of Games, Volume II* (H. W. Kuhn and A. W. Tucker, eds.), pp. 307–317, Princeton University Press, 1953.

[28] S. Babaeinejadsarookolae, A. Birchfield, R. D. Christie, C. Coffrin, C. DeMarco, R. Diao, *et al.*, "The power grid library for benchmarking ac optimal power flow algorithms," *arXiv preprint arXiv:1908.02788*, 2019.

[29] C. Grigg, P. Wong, P. Albrecht, R. Allan, M. Bhavaraju, R. Billinton, Q. Chen, C. Fong, S. Haddad, S. Kuruganty, *et al.*, "The IEEE reliability test system-1996," *IEEE Transactions on Power Systems*, vol. 14, no. 3, pp. 1010–1020, 1999.

[30] H. R. Künsch, "The jackknife and the bootstrap for general stationary observations," *The Annals of Statistics*, vol. 17, no. 3, pp. 1217–1241, 1989.

[31] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.